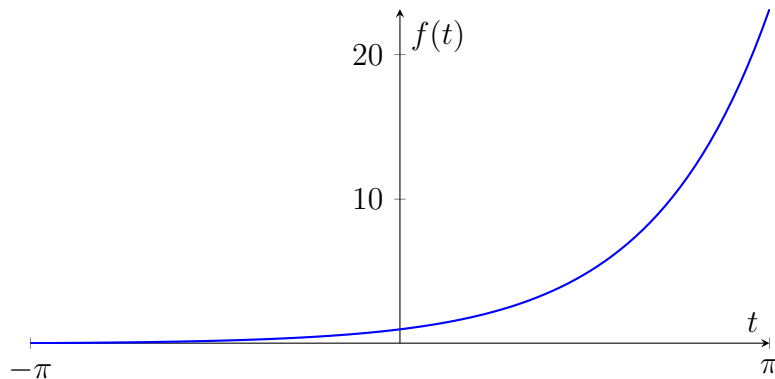


ECE 3210 hw04

1. Find the complex Fourier series of the 2π -periodic signal defined by $f(t) = e^t$ for $-\pi \leq t < \pi$.

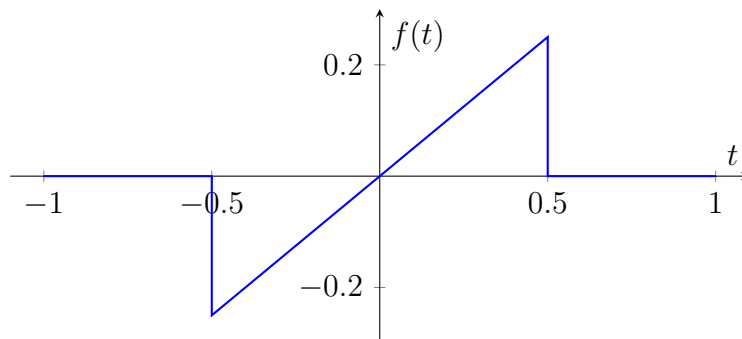


- (a) Find the fundamental frequency ω_0 .
- (b) Find the complex Fourier coefficients D_n in simplest form.
- (c) Represent the Fourier series in terms of D_n .

2. A 2-periodic signal has one period described by

$$f(t) = \begin{cases} t/2, & -1/2 \leq t < 1/2, \\ 0, & 1/2 \leq |t| < 1. \end{cases}$$

Find its complex Fourier series in simplest form.



- (a) Find ω_0 .
- (b) Find D_0 and D_n for $n \neq 0$.
- (c) Represent the Fourier series $f(t)$ in terms of D_n .

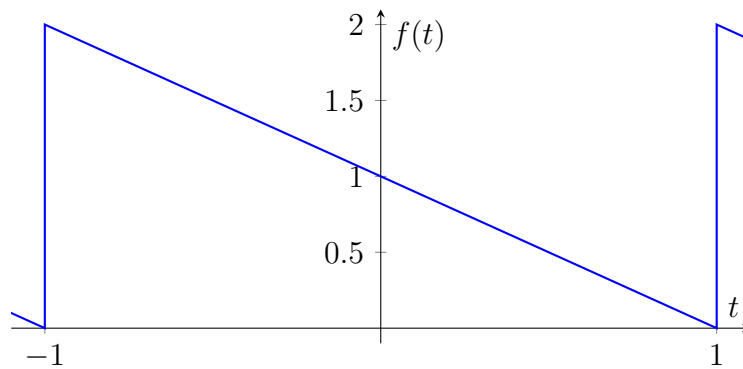
3. Consider

$$f(t) = 0.5 + 5 \sin(0.75t) + 2 \cos(t) + 7 \cos(-0.5t) + \sin(-0.25t).$$

Write the trigonometric Fourier series using fundamental frequency ω_0 and determine the requested coefficients.

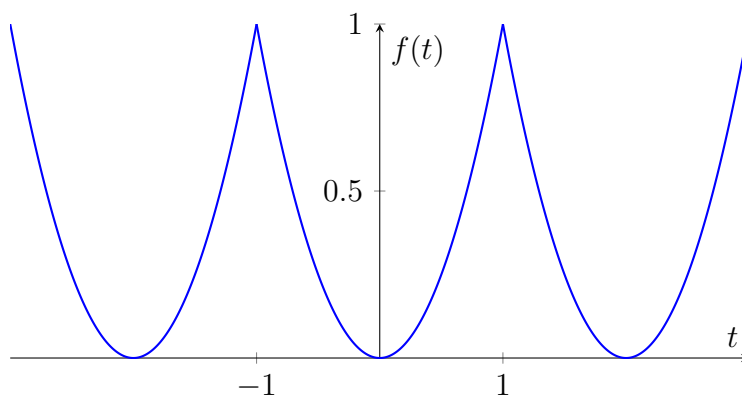
- (a) Find the fundamental frequency ω_0 .
- (b) Find a_0 .
- (c) Find $(a_1, a_2, a_3, a_4, a_5)$.
- (d) Find $(b_1, b_2, b_3, b_4, b_5)$.

4. Find the complex Fourier series of the 2-periodic sawtooth signal defined on $(-1, 1)$ by $f(t) = 1 - t$.



- (a) Find ω_0 .
- (b) Find D_0 and D_n for $n \neq 0$.
- (c) Represent $f(t)$ as a complex Fourier series.

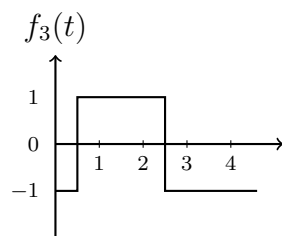
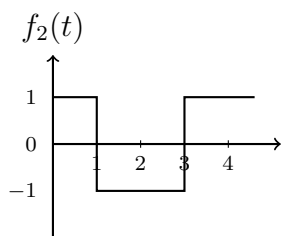
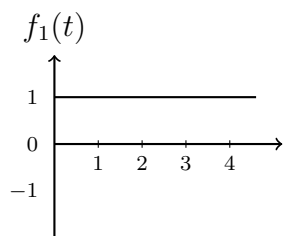
5. Let $f(t)$ be periodic with period 2, and let $f(t) = t^2$ on $-1 \leq t \leq 1$.



- (a) Find the period and fundamental frequency.
- (b) Find the complex Fourier coefficients.
- (c) Represent $f(t)$ as a complex Fourier series.
- (d) For the system with transfer function $H(s) = \frac{s}{s+2}$, find the system response to periodic input $f(t)$.

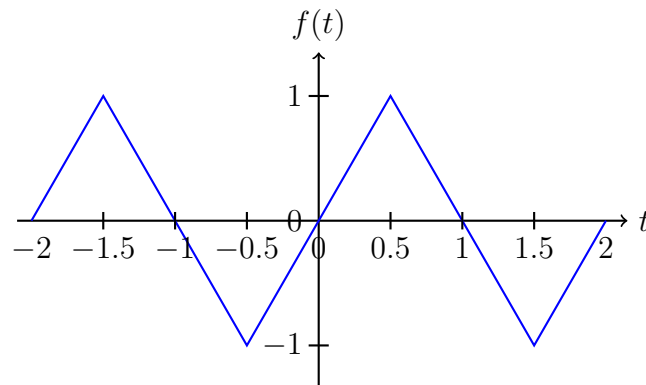
6. Answer each of the following questions about orthogonality.

(a) True or false: the following functions $f_1(t)$, $f_2(t)$, and $f_3(t)$ are mutually orthogonal.



- (b) True or false: the polynomials $f_1(t) = t$, $f_2(t) = t^2$, and $f_3(t) = t^3$ are mutually orthogonal on $t \in [-1, 1]$.
- (c) True or false: the functions $f_1(t) = t \cos(t)$ and $f_2(t) = t \sin(t)$ are orthogonal on $t \in [0, 2\pi]$.

7. Find the trigonometric Fourier series of the function in the figure.



Please reduce your expression to its simplest form.

- (a) Find the fundamental frequency ω_0 .
- (b) Find the value a_0 .
- (c) Find an expression for a_n .
- (d) Find an expression for b_n .
- (e) Represent $f(t)$ as a trigonometric Fourier series.

Answer Key

1(a) $\omega_0 = 1$

1(b) $D_n = \frac{\sinh(\pi)}{\pi} \frac{(-1)^n}{1 - jn}$

1(c) $f(t) = \sum_{n=-\infty}^{\infty} \frac{(-1)^n \sinh(\pi)}{\pi(1 - jn)} e^{jnt}$

2(a) $\omega_0 = \pi$

2(b) $D_0 = 0, D_n = \frac{j}{2\pi n} \left[\cos\left(\frac{\pi n}{2}\right) - \frac{2}{\pi n} \sin\left(\frac{\pi n}{2}\right) \right]$

3(a) $\omega_0 = 0.25$

3(b) $a_0 = 0.5$

3(c) $(a_1, a_2, a_3, a_4, a_5) = (0, 7, 0, 2, 0)$

3(d) $(b_1, b_2, b_3, b_4, b_5) = (-1, 0, 5, 0, 0)$

4(a) $\omega_0 = \pi$

4(b) $D_0 = 1, D_n = (-1)^n \frac{1}{j\pi n}$

4(c) $f(t) = 1 + \sum_{n \neq 0} (-1)^n \frac{1}{j\pi n} e^{jn\pi t}$

5(a) $T_0 = 2, \omega_0 = \pi$

5(b) $D_0 = \frac{1}{3}, D_n = \frac{2(-1)^n}{n^2\pi^2}$ for $n \neq 0$

5(c) $f(t) = \frac{1}{3} + \sum_{n \neq 0} \frac{2(-1)^n}{n^2\pi^2} e^{jn\pi t}$

5(d) $y(t) = \sum_{n \neq 0} \frac{2j(-1)^n}{\pi n(2 + jn\pi)} e^{jn\pi t}$

6(a) False

6(b) False

6(c) False

7(a) $\omega_0 = \pi$

7(b) $a_0 = 0$

7(c) $a_n = 0$

$$7(d) \quad b_n = \frac{8}{n^2\pi^2} \sin\left(\frac{n\pi}{2}\right)$$

$$7(e) \quad f(t) = \sum_{n=1}^{\infty} \frac{8}{n^2\pi^2} \sin\left(\frac{n\pi}{2}\right) \sin(n\pi t)$$